

PORTFOLIO

SUMMARY

Informatics Engineering Student with hands-on experience in data analysis, machine learning, and visualization using Python, SQL, and Tableau. Built forecasting and analytics projects using financial and esports datasets to uncover trends, evaluate performance, and generate data-driven insights.

PORTFOLIO OVERVIEW

Objective: To showcase the ability to analyze and visualize data for impactful and meaningful decision-making.

Focus Areas: Machine Learning · Predictive Modeling · Data Analytics · Data Visualization

FEATURED PROJECTS

Project Name: LQ45 Stock Price Prediction: LSTM vs TFT

Date: 2026

Objective: To compare the performance of LSTM and Temporal Fusion Transformer (TFT) in predicting closing stock prices of 5 LQ45 companies listed on the Indonesia Stock Exchange.

Description:

1. Collected historical stock data via yfinance, stored in MySQL
2. Performed EDA, data cleaning, and feature engineering
3. Normalized data using MinMaxScaler
4. Built LSTM model (evaluated per stock) and TFT model (evaluated aggregate)
5. Compared both models using MAPE, RMSE, MAE, and R^2

Tools & Technologies: Python · SQL · MySQL · yfinance · pandas · numpy · scikit-learn · TensorFlow/Keras · PyTorch · PyTorch Forecasting · matplotlib · seaborn · Google Colab

Outcome: TFT outperformed LSTM based on MAPE (7.45% vs 10.96%) and R^2 (0.9695 vs 0.9108), making it the more reliable model for stock price forecasting on the LQ45 index.

Visuals:

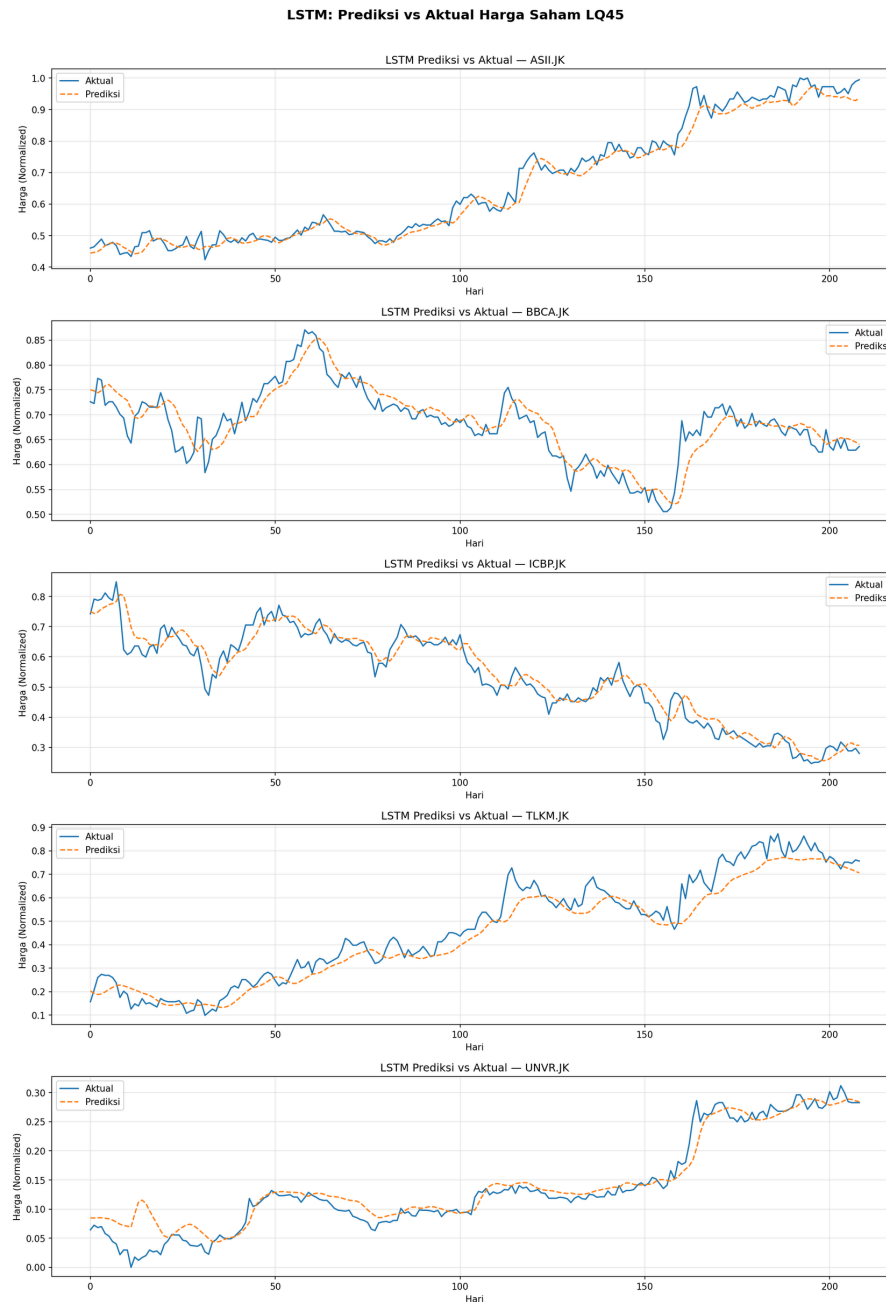


Figure 1. LSTM Actual vs Predicted

Github: <https://github.com/chokiarmando-lab/dota2-team-performance-analysis>

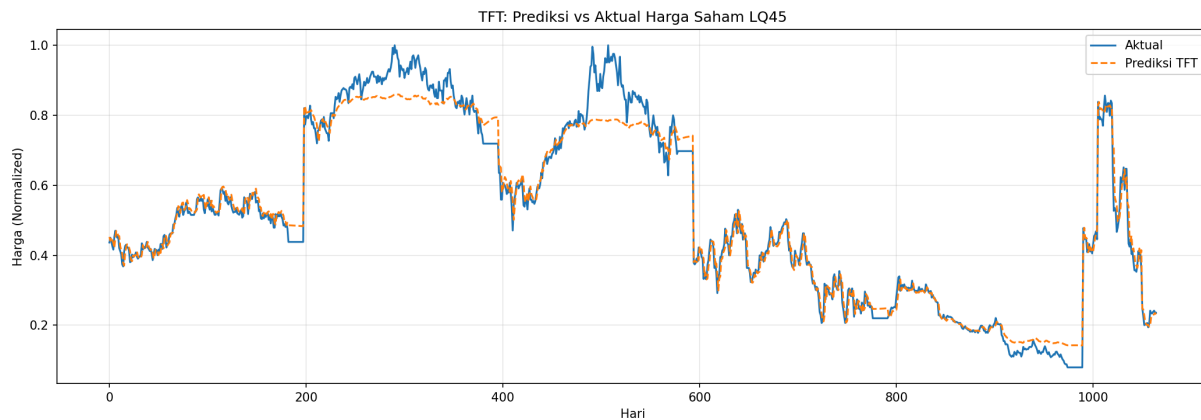


Figure 2. TFT Actual vs Predicted

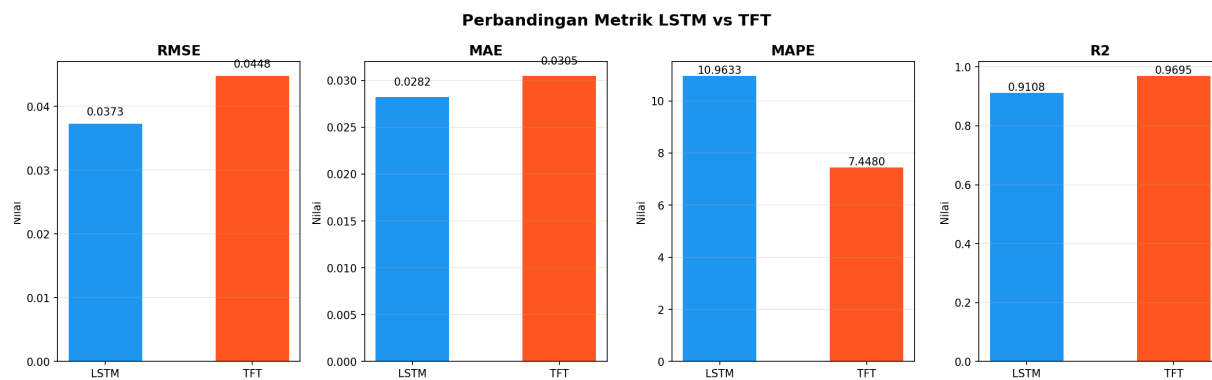


Figure 3. Comparison Metics

Project Name: Dota 2 Team Performance Analysis

Date: 2026

Objective: To analyze professional Dota 2 match data to uncover team performance patterns, hero effectiveness, and mental resilience metrics across 7 major tournaments.

Description:

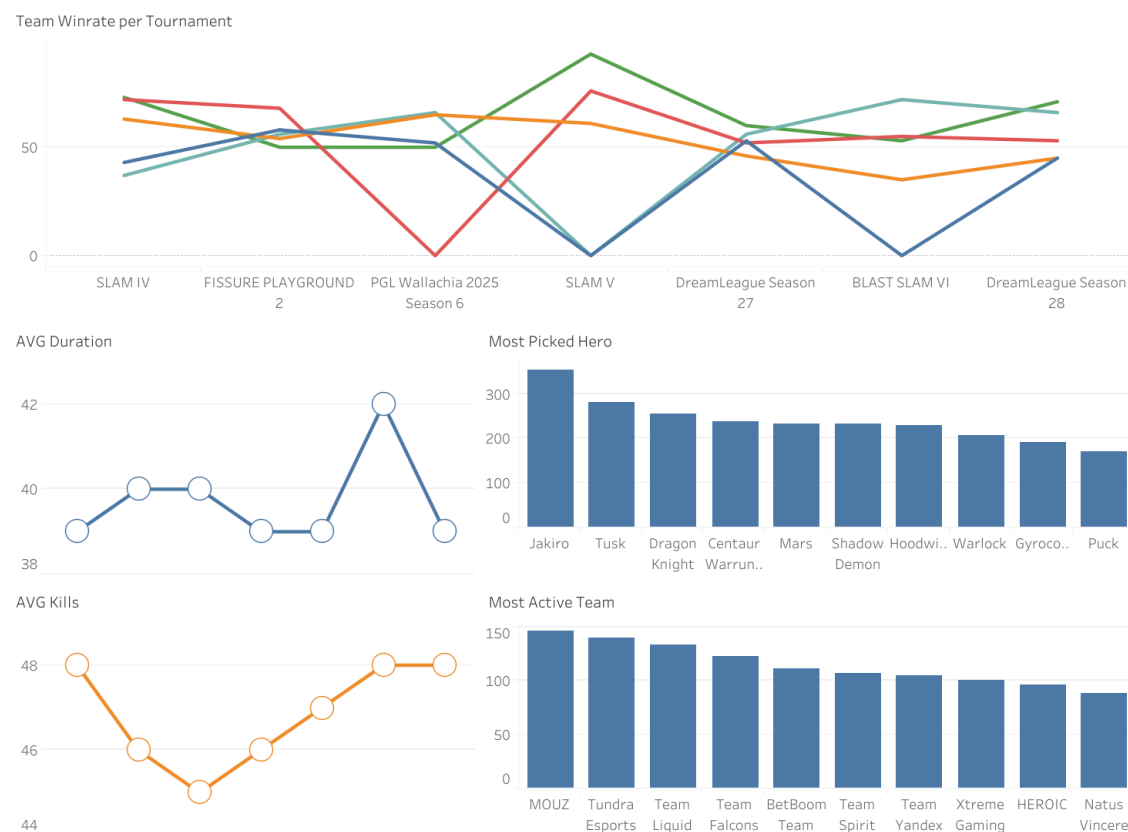
1. Collected match data via OpenDota API covering match results, hero picks, team info, and tournament stats
2. Performed data cleaning and feature engineering
3. Analyzed team performance — win rate, activity level, and aggression metrics
4. Compared hero pick rate vs win rate to identify genuinely effective heroes
5. Built a custom mental score metric (comeback rate – throw rate) to measure team resilience
6. Analyzed match pace and kill trends across tournaments

Tools & Technologies: Python · pandas · matplotlib · seaborn · Plotly · OpenDota API · Tableau

Outcome: Tundra emerged as the most consistent team across all 7 tournaments. Centaur Warrunner stood out as the most reliable hero — ranking 5th in pick rate while maintaining a top 3 win rate. Natus Vincere led the comeback rate, proving the strongest mental resilience in the dataset. Match trends also revealed a shift toward more aggressive gameplay in recent tournaments.

Visual:

Dota 2 Tournament Performance Dashboard



Github: <https://github.com/chokiarmando-lab/dota2-team-performance-analysis>

SKILL HIGHLIGHT

Programming & Query: Python · SQL

Data Processing pandas: numpy · scikit-learn

Machine Learning & Deep Learning: TensorFlow · Keras · PyTorch ·
PyTorch Forecasting

Visualization: matplotlib · seaborn · Plotly · Tableau

Database & Storage: MySQL · Google Drive

Others: Google Colab · yfinance · OpenDota API